



MECH MAINT- BOILER Department
SIMHADRI



Ref:SMTTPP/MECH MAINT-/2022-23/LMI/247903

Date:05/07/2022(04:24:44)

Subject:Review of LMI “HIGH PRESSURE PIPE SUPPORT SYSTEMS”

OPERATION & MAINTENANCE

MECHANICAL MAINTENANCE

(PRESSURE PARTS)

REF.: SMPP/O&M/MM/PP/

Date: 05.07.2022

Subject: Review of LMI “HIGH PRESSURE PIPE SUPPORT SYSTEMS”

LMI No: LMI/OD/OPS/MECH/002

The above approved LMI was prepared in the month of FEB.’2021. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

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On:05/07/2022(04:24:44) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/247903

Few changes have been incorporated in subject LMI as per details mentioned in sl. no-6 of revision index .
Approval may please be accorded for incorporating these changes.

Submitted by:

सुरेश बाबु कोल्ला - उप महाप्रबंधक

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On:06/07/2022(17:43:32) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/247903

Submitted for approval

Submitted by:

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On:07/07/2022(16:07:05) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/247903

Submitted for approval of competent authority

Submitted by:

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On:11/07/2022(10:02:50) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/247903

May please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:17/07/2022(08:45:40) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/247903

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

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On:17/07/2022(10:30:38) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/247903

Approved & Submitted by:

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On:18/07/2022(12:06:50) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/247903

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

LMI/OD/OPS/MECH/002/

**ISSUE NO.1; REVISION NO. 6.00; DATE:
JULY'2022**

**HIGH PRESSURE PIPE
SUPPORT SYSTEMS**

APPROVED FOR

Implementation.....

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	HIGH PRESSURE PIPE SUPPORT SYSTEMS
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OD/OPS/MECH/002/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-1; REV.-6.00; JULY 22</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OD/OPS/MECH/002</u>
<u>4.PREPARED BY:</u> Sr. MANAGER (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY:</u> DGM(MM-BMD) AGM (BMD) GM (MTC.) GM (O&M)	SIGNATURE----- SIGNATURE_____ SIGNATURE_____ SIGNATURE_____
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE_____ HOP, SIMHADRI
<u>7.DATE OF ISSUE</u>	
<u>8.INDEX LIST</u>	ENCLOSED.
<u>10.DISTRIBUTION LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: HIGH PRESSURE PIPE SUPPORT SYSTEMS REV: 6.00		
DATE: JULY22		
REF NO. LMI/OD/OPS/MECH/002/		PAGE : 1 OF 8
<u>1.0 INDEX</u>		
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2	INTRODUCTION	2
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NTPC	LMI	SIMHADRI
TITLE: HIGH PRESSURE PIPE SUPPORT SYSTEMS REV: 6.00 DATE: JULY 22		
REF NO. LMI/OD/OPS/MECH/002/		PAGE : 2 OF 8
<u>2.0 INTRODUCTION</u> High pressure piping (carrying fluids of pressures greater than 30 bar) are supported by constant load hangers, variable spring load, rigid struts & sliding supports. Even though these support units i.e. constant / variable load have limited maintenance, thermal loading can exert significant additional forces above or below the design limits on the pipe systems which can in-turn produce expansion/ contraction. They in-turn induce high stresses which may lead to weld crack or deflection problems at the boiler headers or turbine casings. The objective of this LMI is to establish system of periodic preventive checking and associated maintenance for high pressure pipe support systems. <u>3.0 SCOPE</u> The scope of this LMI is to detail the methods that are to be adopted for the periodic inspection and maintenance of high pressure pipe support systems. It covers all high pressure piping viz. MS, CRH, HRH, feed water & HP bypass. <u>4.0 REFERENCE DOCUMENT</u> Operation directive OD/OPS/MECH/002 circulated by operation services - NTPC corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> HIGH PRESSURE PIPE SUPPORT SYSTEMS REV: 2.00 DATE:APR'2013 <u>6.0 SPECIFIC INSTRUCTIONS</u> For sustained integrity of high pressure pipe support systems the following instructions are to be followed - Periodic inspection of hanger supports. - Comparison of hot and cold setting with the design settings and suitable corrective adjustment. Colour coding for hot and cold reading marking on hanger scale to be done. - Inspection of hanger support of the particular piping in which there is an occurrence of abnormal hammering. RLA and stress analysis to be carried for units > 25 years of age.		

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<u>7.0 METHODOLOGY OF IMPLEMENTATION & RESPONSIBILITY CENTERS</u>		
a) Pre-overhaul survey of high pressure pipe support systems to be carried out at unit full load condition. This survey would include checking of integrity of supports and scantlings in their ‘as found’ movement condition, recording of spring scale reading and mechanical condition and checking for evidence of gross clip distortion, pipe work fouling and any abnormal operational feature. Provision of permanent and safe approaches for CD bay and other nonreachable locations to be made. Responsibility – HOS (PP) & HOS (TMD) Frequency – Once before every overhaul		
b) Survey to be carried during overhaul along with BHEL/Ex-BHEL experts. This would include checking of all constant and variable load hangers, restraints and sliding supports and recording of cold setting values and mechanical condition. Responsibility – HOS (PP) & HOS (TMD) Frequency – During every overhaul		
c) Comparison of survey results is to be carried out with design settings. All the defects are to be listed out and corrective action is to be taken to attain original condition including replacement if necessary. Responsibility – HOS (PP) & HOS (TMD) Frequency – During every overhaul		
d) In case of abnormal hammering all the supports, constraints of the particular piping shall be physically inspected against snapping and distortions. Taking corrective action would include replacing damaged supports, lifting of piping and reloading of supports, recording of cold set values and full load hot values after any rectification and comparison with design settings. Responsibility – HOS (PP) & HOS (TMD)		
e) Post-overhaul survey to be carried out at full load condition. This survey would include checking of integrity of supports and scantlings in their ‘as found’ movement condition, recording of spring scale reading and mechanical condition and checking for evidence of gross clip distortion, pipe work fouling and any abnormal operational feature. Responsibility – HOS (PP) & HOS (TMD) Frequency – After every overhaul		
f) All the pre & post-overhaul and overhaul survey results and exceptions are to be sent to corporate engineering and corporate OS. Responsibility – HOS (PP) & HOS (TMD)		

NTPC	LMI	SIMHADRI
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8. EXCEPTION REPORTING

The exceptions encountered during inspection are to be reported as per the following format:

The following are to be treated as significant exceptions

- a) If more than 25% or three adjacent variable spring units have bottomed out.
- b) If the deflection of any limit stop is in the direction opposite to design
- c) Any other case in the opinion of HOP, SIMHADRI.

Description of exception	Unit No.	Location of exception	Support no.	Remarks

9.0 RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to inspection and maintenance of high pressure pipe support systems as detailed in the LMI are with HOS (PP) for boiler side piping & HOS (TMD) for Turbine side piping.

10.0 RECORD KEEPING

The records to be maintained under this LMI include

- a) compliance record of the activities detailed under this LMI
- b) reports of pipe line survey carried out
- c) **Proper record and history of piping hangers to be maintained and making hanger report available in soft form in BMD dashboard/ intranet.**
- d) Exception reporting

The above records are to be maintained by Area Engr. (Pr. Parts) & Area Engr. (TMD)

11.0 REVIEW

This document shall be reviewed by HOP, SIMHADRI on TWO yearly basis.

NTPC		LMI		SIMHADRI
TITLE: HIGH PRESSURE PIPE SUPPORT SYSTEMS REV: 6.00 DATE: JULY'22				
REF NO. LMI/OD/OPS/MECH/002/			PAGE : 5 OF 8	
12.0 REVISION INDEX(for changes/revision)				
Sl. No.	Rev. no.	Date	Details	Approved by
01	001		i) Approval routing changedat “LMI APPROVAL DOCUMENT”	
			ii) Responsibility changed from SS (PP) to DGM (PP)& DGM (TMD) at Responsibilities	
			iii) Reviewing authority changed from DGM (MM) to GM (I/c)	
			iv) Distribution List	
02	002	30.04.13	i) Responsibility changed from DGM (PP) & DGM (TMD) to HOS (PP) & HOS (TMD) at Responsibilities	
			ii) Inclusion of Unit#3&4 in compliance record	
			iii) Distribution List	
03	003	31.05.16	i) Approval routing changed at “LMI APPROVALDOCUMENT”	
04	004	07.01.2020	i) Approval routing changedat “LMI APPROVAL DOCUMENT” ii) Methodology & Implementation Clause 7(b),changed from "BHEL" to "BHEL/Ex-BHEL" iii) Responsibility centers changed from "Mech Mtc." to HOS (PP) & HOS (TMD) iv) Review Frequency changed from "three years"to "TWO years" iv) Distribution List	
05	005	19.02.2021	According to LMI 3rd Party Audit Observations	

			conducted from date 24.09.2020 to 01.10.2020 comparison of pipe support measurements is against erection readings as per LMI. But practice is comparison against design values. Changed in LMI accordingly	
06	006	05.07.2022	i) Proper record and history of Piping hangers to be maintained, has been incorporated. ii) Making hanger report available in soft form in BMD dashboard/intranet incorporated. iii) Colour coding for Hot & Cold reading marking on hanger scale incorporated. iv) Provision of permanent & Safe approaches to be made for hangers in CD bay and other non reachable locations. v) Guidelines for Piping Hangers checking in Boiler > 25 Years age has been incorporated vi) Distribution list	

[illegible]

NTPC	14. COMPLIANCE RECORD	SIMHADRI
TITLE: HIGH PRESSURE PIPE SUPPORT SYSTEMS REV: 5.00		
DATE: FEB'21		
REF NO. LMI/OD/OPS/MECH/002/		PAGE : 7 OF 8
UNIT NO.	ACTIVITY	COMPLIANCE
I/II/III/IV		

[illegible]